Calibration: β

$$(1+i^{A}) = (1+\pi^{e})\left(\frac{1}{\beta}\right) \qquad \boxed{\frac{1}{\beta} = \frac{1+i^{A}}{1+\pi^{e}}}$$

- β is the discount factor.
- In steady-state we have $(1+i)/(1+\pi) = 1/\beta$. Hence, this parameter determines the long-run real interest rate in the economy.
- We want to choose β to target a reasonable real interest rate.
- For example, annual real interest rate of 2% means $\beta = 1/1.02$ if our model period is a year, and $\beta = (1/1.02)^4$ if our model period is a quarter.

i: Three month US T-Bill

Alesina and Summers (1993)

- Measure central bank independence based on factors such as:
  1. Are Governor and Board appointed by the government?
  2. What is the length of the appointments?
  3. Are there government representatives on the board?
  4. Is government approval for monetary policy decisions required?
  5. Is price stability objective explicit ?
  6. How easy is it for government to finance deficits with Central Bank credit?
- How does Central Bank independence correlate with macroeconomic factors?

Alesina and Summers (1993): Average inflation

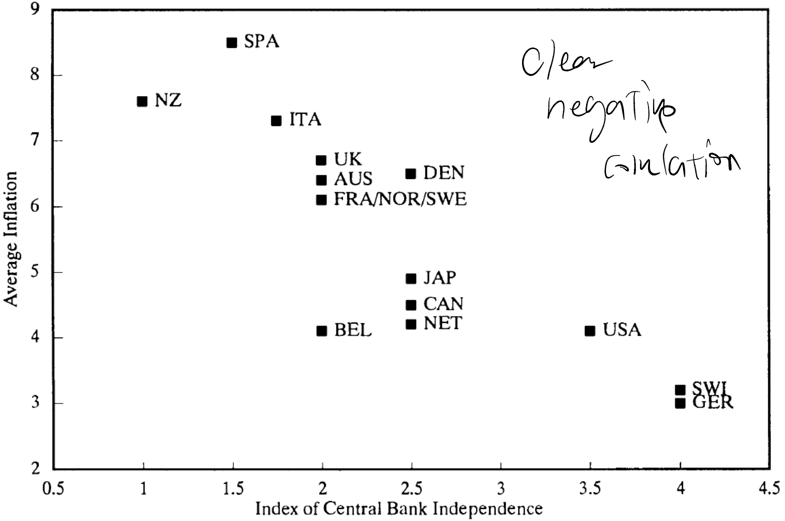


FIG. 1a. Average Inflation

Alesina and Summers (1993): Variance of inflation

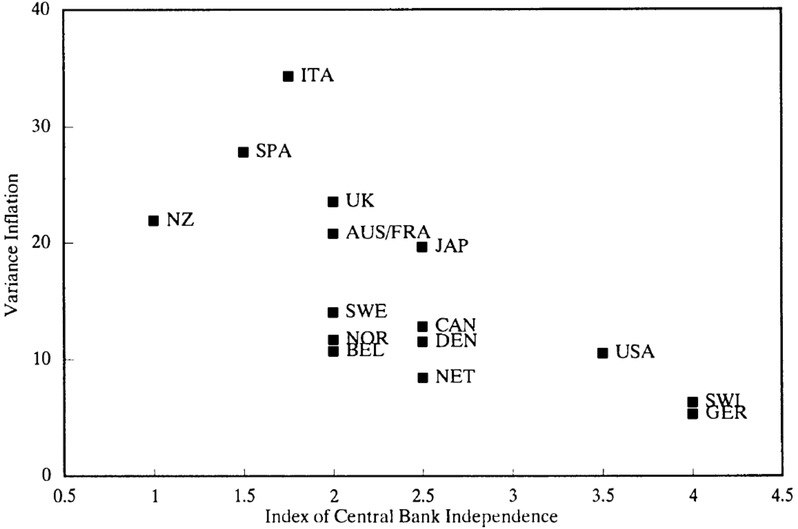


FIG. 1b. Variance Inflation

Alesina and Summers (1993): Average GDP growth

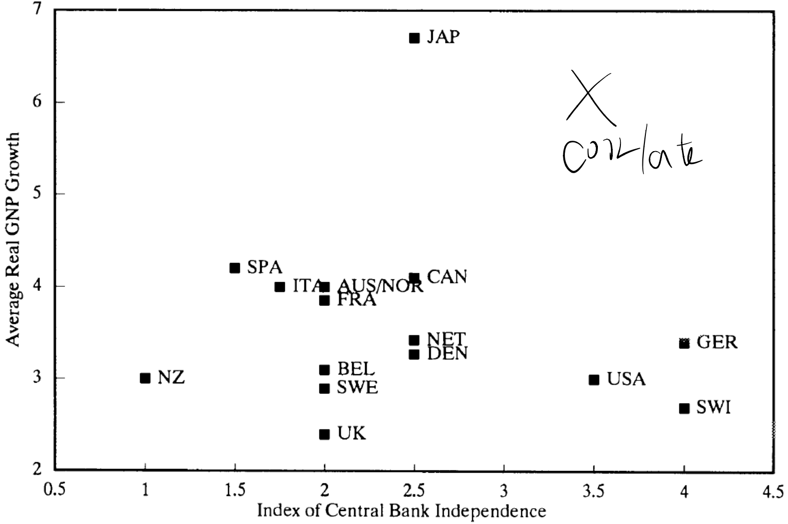


FIG. 2a. Average Real GNP Growth

Central Bank independence and output growth

- In a different paper Cukierman et al. (1993) conclude:

  "A main finding of the paper is that Central Bank Independence has a positive effect on growth in LDCs (less-developed countries) and no effect within industrial countries."

Bond pricing with different maturities

- With no uncertainty, the math shows that:

  1. $(1 + i_t^L) = (1 + i_t^S)(1 + i_{t+1}^S)$

  No-arbitrage means that buying a two-period bond should be the same as buying the one-period bond and then reinvesting it the following period.

  ? inflation → ? interest rate tmr

- With uncertainty this result need not hold because the two investments are no longer the same. The two-period bond yields a fixed nominal return while the one-period bond reinvestment strategy does not because future nominal interest rates can vary.

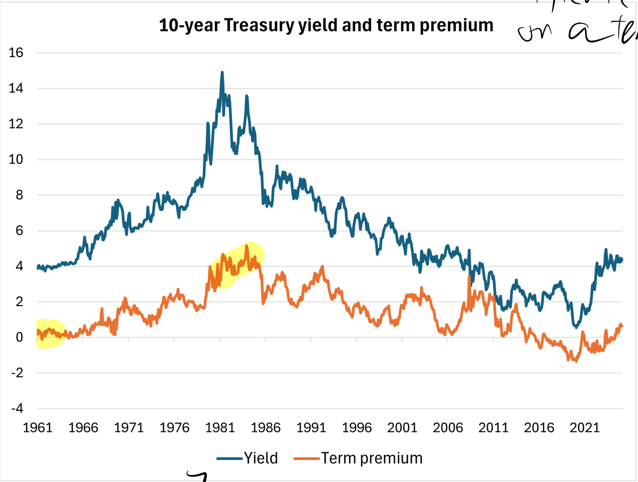
- Difference between $(1 + i_t^L)$ and $\mathbb{E}_t\left[(1 + i_t^S)(1 + i_{t+1}^S)\right]$ is called the term premium.

typically positive

2-period Bond → Compensation for hold Longer term Bond

Estimates of term premium in US

interest rate on a ten year bond



+ve

$\Rightarrow (1+i_t^{10}) > \mathbb{E}[(1+i_t^{s}) \cdots (1+i_{t+1}^{s}) \cdots]$

Figure: Source: NY Fed

Calibration: what it is

- When we have a quantitative model we want to use it to get quantitative answers.

- Calibration consists of using data to choose the parameters of our model, so that we can then use it to get these quantitative answers.

- In our case, we are interested in measuring the welfare costs of inflation. So we need to find ways of choosing the several parameters, so that we can compute $\gamma(\pi)$ as a number.